FinQuest India – Mid-Submission

Megathon X Vanguard India | Problem Statement: Empowering the Next Generation Through Financial Literacy

Team: [Team Name] | Members: [Names] | Institution: [College] | Date: [Submission date]

1. Abstract

FinQuest India is a mobile-first, game-based financial-literacy web platform for Grade 11–12 students, especially those from underserved communities in Hyderabad, and it stays friendly for anyone aged 12 and above. Instead of articles and definitions, learners learn by doing: they make a choice, see the consequence, and try again in a safe, fictional environment.

The platform covers the four required pillars: **Plan & Budget, Save & Manage, Understand Investing, and Stay Financially Safe**. Each concept is taught through relatable rupee-based scenarios, interactive experiments, mini-games and scenario-based quizzes. A persona selector adapts examples and tone by age and financial maturity. The full learning experience is available in English and Telugu, with a structure ready for Hindi and other languages. A privacy-first design means no login or personal data is needed to learn, and an NGO/educator view supports rollout across about 1,500 students and beyond.

We follow the official guardrail throughout: **educate, do not recommend investments; simulations are assumptions, not promises.**

2. Problem Understanding

- Many students from underserved communities have little exposure to budgeting, saving, investing or fraud safety, and learn about money late or through mistakes.
- Existing content is mostly text, video or definitions, which is easy to forget and hard to apply.
- Students need a safe place to experiment, make mistakes and build confidence without real-world consequences.
- NGOs need a simple way to deliver, roll out and track progress without collecting sensitive student data.

3. Our Solution in One View

PS expectation	How FinQuest delivers it
Simple + relatable	Everyday ₹ scenarios (pocket money, phone repair, festival spending, UPI requests); plain language first, technical term second
Interactive + fun	10+ mini-games, XP, levels, streaks, badges, a journey map and a mascot guide
Safe to explore	Fictional money and companies only; clearly labelled simulations; "Try again" feedback
Persona-led	Age/context selector (12–17, 18–25, 26–40, 41–60, 60+), with 12–17 as the default
Language-ready	English and Telugu for the whole learning experience; Hindi-ready i18n structure
Game-based assessment	Scenario quizzes, boss levels and mastery checks with targeted remediation
Intuitive UI	Bottom tab bar, big tap targets, minimal text, a visual path to follow
NGO-integrated	Class code, aggregate completion view, printable reports (sample data clearly labelled)
Privacy-conscious	No login, no PII, local-only progress, a Reset-my-data button
Scalable	Static/edge-friendly front end, content-driven design, no heavy backend needed for MVP

4. Target Users and Personas

- **Primary:** Grade 11–12 students in Hyderabad, including those in NGO-supported programmes.
- **Secondary:** Students aged 12+, first-time earners, parents and senior citizens (lower priority; reuse the same engine with persona-specific content).
- **Supporting:** NGO coordinators and teachers who deliver and monitor the programme.

5. Learning Approach

Every concept follows a flexible teaching flow, adapted to its difficulty rather than forced into identical screens:

1. Hook: a relatable question or dilemma.
2. Predict: the learner guesses before seeing the answer.
3. Visual explanation of the idea.
4. Step-by-step mechanism and a worked example.
5. Interactive experiment where the learner changes variables.
6. Counterexample or comparison.
7. Feedback that explains why an answer is right or wrong.
8. Misconception buster (for example, "a ₹10 NAV fund is cheaper than a ₹100 NAV fund" is false).
9. Practice on a new scenario.
10. Mastery check, then targeted remediation and later revision.

6. Curriculum Coverage (Four Pillars)

Pillar 1: Plan & Budget – money basics, needs vs wants, 50-30-20 thinking, opportunity cost, SMART goals, handling surprise expenses.

Pillar 2: Save & Manage – savings accounts, emergency fund, FD, RD, simple vs compound interest, inflation, real vs nominal return, EMI basics, deposit protection.

Pillar 3: Understand Investing – risk, time, compounding, diversification; shares and stock exchanges, IPO, price discovery, order types, valuation basics (market cap, EPS, P/E), mutual funds (units, NAV, AMC, fund types, active vs passive, costs, direct vs regular), SIP and rupee-cost averaging, CAGR, PPF and NPS awareness. All presented as education and trade-offs, never as recommendations.

Pillar 4: Stay Financially Safe – OTP and PIN protection, QR/collect-request fraud, screen-sharing scams, fake KYC, Ponzi and pump-and-dump schemes, fake trading apps, fake advisors, how to verify registrations, and where to report (cybercrime helpline, SEBI SCORES).

7. Mini-Games and Quizzes

Game	Pillar	What the learner does and discovers
Needs vs Wants Swipe	1	Swipes everyday items into buckets; sees a live pie chart
Budget Survival	1, 2	Manages ₹15,000/month through surprise expenses; sees why an emergency fund matters
Compound Growth Garden	3	Grows a "money tree" and compares simple vs compound growth; Rule of 72
Inflation Shopper	2	Moves a years slider and sees purchasing power shrink
NAV Builder	3	Changes fictional company prices and watches fund NAV recalculate
SIP Time Machine	3	Edits monthly NAVs; sees units bought, average cost and value
Price Match	3	Places buy/sell orders in a fictional order book and sees price discovery
Market Mayhem	3	Virtual ₹1,00,000, fictional companies, 10 rounds of news; learns diversification
Scam or Safe	4	Inspects fictional messages and chooses a response before seeing the red flags
UPI Defender	4	Timed Approve / Block / Report decisions
Daily 5 Quiz and Boss Levels	All	Scenario questions; drives streaks and mastery

Rewards: XP, levels (Beginner to Master), daily streak, badges (Budget Boss, Compound Wizard, SIP Scholar, Scam Slayer, FinQuest Master) and shareable result cards that carry no personal data.

8. Skeleton Structure (Site Map)

```
FINQUEST INDIA
├── 1. Home
|   ├── Welcome to FinQuest
|   ├── Choose Your Financial Journey (persona picker)
|   ├── Explore Financial Goals
|   ├── Learn About Money
|   ├── Compare Savings & Investments
|   ├── Financial Calculators
|   └── Start My Financial Plan
├── 2. My Financial Journey (persona-led)
|   ├── Ages 12–17: Money Basics, Saving & Budgeting, Banking & Digital
Safety,
|   |             Virtual Money Simulator, Financial Skills & Quizzes
|   ├── Ages 18–25: College Budget Planner, First Income Planning, Emergency
Savings,
|   |             Banking & Credit Basics, Investment Fundamentals
|   ├── Ages 26–40: Salary & Expense Planning, Emergency Fund, House &
Vehicle Goals,
|   |             Family Financial Planning, Long-Term Investing Education
|   ├── Ages 41–60: Family Financial Security, Children's Education Planning,
|   |             Loan & Debt Management, Retirement Preparation, Investment
Risk Review
|   └── Ages 60+:  Retirement Income Planning, Essential Expense Budget,
Savings & Deposit
|                 Education, Healthcare Expense Planning, Inflation &
Longevity, Fraud Protection
├── 3. Savings & Investment Explorer
|   └── Savings Accounts, FD, RD, Post Office & Govt Schemes, Bonds, Mutual
Funds,
|       Index Funds & ETFs, Shares & Stock Market, Pension Options, Insurance
Education
├── 4. Which Option Suits My Goal? (trade-off explorer, no recommendations)
|   └── Emergency, Education, Short-Term Purchases, House & Vehicle, Family
Protection,
|       Long-Term Wealth Building, Retirement Income
├── 5. My Money Planner
|   └── Income & Expense Tracker, Monthly Budget, Savings Goals, Emergency
Fund Planner,
|       Debt & EMI Planner, Inflation Calculator, Monthly Financial Report
├── 6. Life Goals & Retirement
|   └── First Laptop or Vehicle, Higher Education, Marriage & Family, Home
```

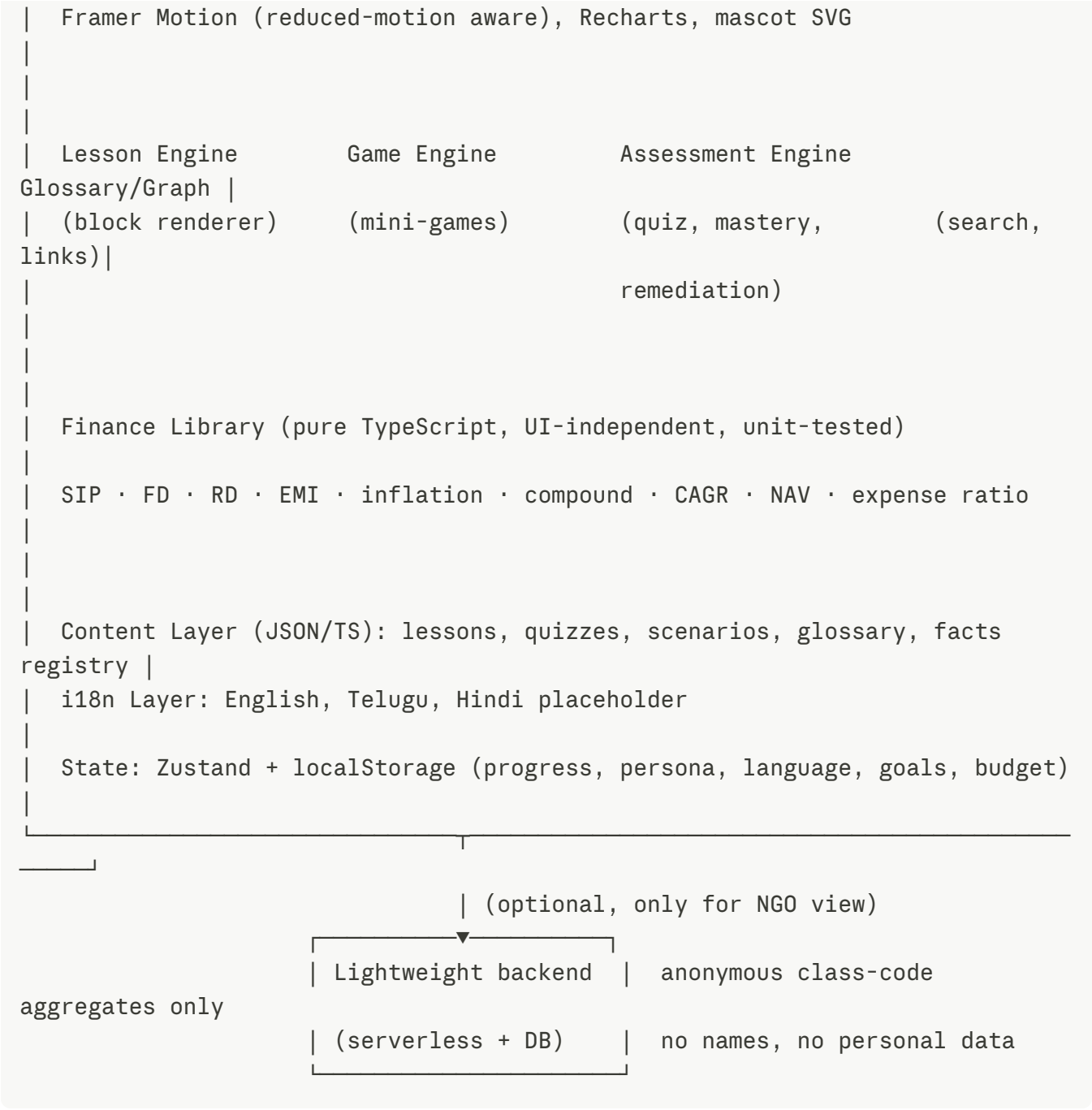
```
Purchase,
|       Children's Education, Financial Independence, Retirement Planner
├── 7. Financial Learning Academy
|   └── Beginner Lessons, Budgeting & Saving, Banking & Credit, Investing &
Risk,
|       Taxes & Fees, Financial Scams, Interactive Quizzes, Virtual Market
Simulator
├── 8. Calculators & Comparisons
|   └── Budget, FD & RD, Savings Goal, SIP Illustration, Loan EMI,
Retirement,
|       Inflation, Investment Comparison
├── 9. My Financial Dashboard
|   └── Goals, Savings Progress, Learning Progress, Financial Health
Checklist,
|       Progress Charts, Download My Plan
├── 10. Trust, Safety & Support
|   └── Privacy & Data Protection, Age-Appropriate Experiences,
Parent/Guardian Info,
|       Education Disclaimer, Verified Official Resources, Help & Feedback
├── Glossary (50+ searchable terms linked to lessons)
└── Teacher / NGO View (class code, aggregate progress, printable report)
```

9. Student Journey (User Flow)

1. Open the app (no login) and pick a language and persona.
2. See the journey map with nodes for the four pillars.
3. Open a lesson: hook, predict, visual, experiment, feedback.
4. Play the linked mini-game and earn XP and badges.
5. Take the scenario quiz; weak areas trigger a targeted revision card.
6. Use calculators and the money planner for personal what-if experiments.
7. View the dashboard, download the plan, and (with a class code) share aggregate progress with the NGO.

10. System Architecture

```
┌──────────────────────────────── Client (Browser / PWA)
──────────────────────────────┐
|  Presentation Layer: Next.js App Router pages, Tailwind + shadcn/ui,
|
```



Why no mandatory backend for MVP: learning, games and calculators run entirely in the browser, which keeps the product privacy-conscious, cheap, fast on low bandwidth and easy to scale. A small serverless backend is added only for anonymous NGO aggregates.

11. Technology Stack

Area	Choice	Reason
Framework	Next.js 14+ (App Router) with TypeScript	Routing, performance, easy deployment
Styling	Tailwind CSS + shadcn/ui	Fast, consistent, accessible components
Animation	Framer Motion	Playful feedback, respects reduced motion
Charts	Recharts	Interactive financial charts with text alternatives
State	Zustand + localStorage	Simple, no login, privacy-friendly
i18n	JSON message files (next-intl or equivalent)	Full English/Telugu learning experience
Testing	Vitest (unit), Playwright (smoke)	Verified calculations and flows
Delivery	PWA, lazy loading	Low-bandwidth and offline-friendly
Build tool	Antigravity, phase by phase	Incremental, tested implementation

12. Repository Structure

```
/app                routes: home, journey, explorer, goals, planner,
academy,
                    tools, play, dashboard, glossary, teacher, trust
/components         ui, layout, mascot, charts, lesson-blocks, games
/lib/finance        pure calculation functions (SIP, FD, RD, EMI, CAGR, NAV,
...)
/lib/engine         lesson engine, assessment, remediation, XP and badges
/content            lessons/, quizzes/, games/, glossary/, personas/, facts/
/messages           en.json, te.json, hi.json
/store              Zustand stores and localStorage persistence
/tests              unit (finance), content validation, e2e smoke
/docs               correction-register.md, roadmap.md, source-list.md
```

13. Content Schema (Summary)

Every lesson, quiz item and glossary term is data, not hardcoded UI:

- id, pillar, category, prerequisites, learning objectives, depth (beginner/intermediate/advanced)
- blocks (hook, predict, explain, example, interaction, counterexample, misconception, practice, mastery)
- glossary links and related topics
- source reference, lastReviewed, reviewStatus, timeSensitive flag
- translations keyed by language

This lets one reviewed fact or translation update consistently across lessons, calculators, glossary entries and games.

14. Financial Accuracy and Trust

- All calculations are pure functions with unit tests and edge cases (0% rate, 0 years, large values).
- Every calculator shows its formula, timing convention and rounding assumption.
- Time-sensitive facts (tax rates and thresholds, scheme rates, deposit insurance limit, market rules) live in a facts registry with source URL, as-of date and review date, and stay marked "verify" until checked against official sources.
- Worked check example: SIP of ₹2,000/month at an assumed 12% p.a. for 10 years gives about ₹4,64,678 for start-of-month contributions (₹4,60,077 for end-of-month), with ₹2,40,000 invested. This is shown as an assumption, never a prediction.
- Other verified checks: FD ₹50,000 at 7% for 3 years (quarterly) ≈ ₹61,570; ₹1,00,000 at 6% inflation for 10 years ≈ ₹55,839 of today's purchasing power; CAGR of ₹1,00,000 to ₹1,80,000 over 5 years ≈ 12.47%.
- Mandatory disclaimer on every calculator, simulation and investing page.

15. Accessibility, Language and Privacy

- English and Telugu for the whole learning experience (instructions, examples, feedback, quizzes and games), with unreviewed strings flagged until a fluent reviewer approves them.

- Mobile-first at 390px, large tap targets, keyboard navigation, WCAG AA contrast, text summaries for charts, reduced-motion support, dark mode and low-bandwidth design.
- No login to explore, no personal data collection, no third-party trackers, local-only progress with a Reset-my-data button, and sample data always labelled as sample.

16. NGO and Educator Integration

- Class code lets a coordinator view anonymous aggregate completion and quiz performance.
- Printable progress report and a facilitator guide for group sessions.
- Works on shared or low-end devices and offline after first load.
- Clearly separates sample data from real learner records.

17. Scope: MVP vs Long-Term

Hackathon MVP: persona picker; full journeys for the default 12–17 persona across all four pillars; at least 6 mini-games; 5 or more calculators; glossary; dashboard; Telugu for the core journey; NGO view with sample data.

Long-term: full depth for all five personas, Hindi and more languages, company financial-statement and valuation lessons, advanced return measures (XIRR), adaptive remediation, real NGO analytics with consent, and expert-reviewed content.

Topics not yet built are recorded as planned in the roadmap and are not claimed as complete.

18. Development Roadmap

Phase	Focus
1	Baseline, design system, mascot, navigation, persona and language switcher, correction register
2	Finance library with unit tests and facts registry
3	Content schema, lesson engine, first full journey (Needs vs Wants, Budget Survival, Savings Goal)

Phase	Focus
4	Home, journey map, explorer, goal trade-offs, money planner
5	Calculators with charts and text alternatives
6	Investing lessons with Compound Garden, SIP Time Machine, NAV Builder, Price Match
7	Safety lessons with Scam or Safe and UPI Defender
8	Market Mayhem, Inflation Shopper, Daily 5, boss levels
9	Glossary and concept links, remediation logic
10	Telugu translation, accessibility pass, PWA and performance
11	Dashboard, Download My Plan, Teacher/NGO view
12	Integration testing, fixes, deployment, demo

19. Progress Status (Fill In Before Submitting)

Be honest and specific so judges can trust the status.

- Completed so far: [list the phases or features that are actually built and tested]
- In progress: [list]
- Planned for final submission: [list]
- Known gaps or unverified items: [list]
- Link to prototype or repository: [URL]

20. Risks and Mitigation

Risk	Mitigation
Incorrect financial content	Unit-tested formulas, facts registry with sources and review dates, expert review before launch

Risk	Mitigation
Content seen as investment advice	Guardrail banner, trade-off-only comparisons, no named products
Poor translation quality	Plain-language glossary, native-speaker review flags, English fallback
Low-end devices and slow networks	PWA, lazy loading, light assets, no heavy video
Student privacy	No login, local-only progress, anonymous class-code aggregates only
Scope too large for the timeline	Phased MVP, long-term items recorded as planned

21. Expected Impact and Next Steps

- Students gain practical confidence in budgeting, saving, basic investing concepts and fraud safety through repeated safe practice.
- NGOs get a ready-to-deploy, low-cost, privacy-respecting programme that can start with about 1,500 students and scale.
- Next steps for final submission: complete remaining phases, verify time-sensitive facts against official sources, review the Telugu content with a fluent speaker, run accessibility and device testing, pilot with a small student group, and prepare the three-minute demo.

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